

Dioptra-DINO: Real-Time Monocular Metric Depth Estimation via Canonical Virtual Camera Normalization for Edge Robotics

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Abstract—Monocular metric depth estimation on autonomous mobile robots presents a persistent trade-off between physical scale calibration and inference throughput. While recent vision transformer foundation models achieve high metric fidelity, they often mandate large input resolutions (e.g., 616×1064) and multi-hundred-millisecond latencies, impeding real-time closed-loop robotic control. Conversely, lightweight relative depth estimators suffer from severe scale ambiguity and cannot recover metric distance without test-time oracle alignment or uncalibrated priors. In this work, we present **Dioptra-DINO**, an efficient 27.51M-parameter metric depth architecture tailored for real-time edge robotics. Dioptra-DINO couples a self-supervised DINOv2-Small backbone with a Canonical Virtual Camera transformation ($F_{\text{canon}} = 1000.0\text{px}$), Trivision Ray FiLM Modulation, and an Angular Residual Attention (ARA) module, enabling robust scale invariance directly at a native resolution of 336×336. Across extensive empirical evaluations spanning 3,000 indoor test frames, 21 diverse architectural categories, and head-to-head testing against contemporary camera-conditioned foundation models (Metric3D and UniDepth V2), we observe: (1) On photorealistic ray-traced interiors (Apple Hypersim, 2,744 frames), Dioptra-DINO attains an absolute relative error (**AbsRel**) of **0.1477** and inlier precision (δ_1) of **84.3%**, outperforming Metric3D ViT-Small (AbsRel **0.2259**, δ_1 **73.4%**) by **34.6% in relative error** and UniDepth V2 (**0.2164**, δ_1 **76.2%**) by **31.7% in relative error**. (2) When placed on an identical 336×336 playing field across 100 indoor frames, Dioptra-DINO achieves **0.2290 AbsRel** and **72.5% inliers**, whereas Metric3D collapses to **0.3997 AbsRel** and **17.3% inliers**, revealing that Metric3D’s metric calibration heavily degrades at lower spatial resolutions. (3) Operating at native 336×336, Dioptra-DINO runs in **58.2–62.9 ms** (15.9–17.2 FPS) on Apple Silicon GPU and consumes under 240 MB VRAM, achieving a **12.6× speedup over Metric3D** (753.5 ms) and **6.2× speedup over UniDepth V2** (421.6 ms). We candidly report failure modes, document multi-domain training distributions, and release all benchmark code, evaluation protocols, and fine-tuned checkpoints to foster reproducible edge robotics perception.

I. INTRODUCTION

Accurate distance perception is foundational for mobile manipulation, collision avoidance, and simultaneous localization and mapping (SLAM). While active depth sensors (e.g., LiDAR, time-of-flight, and structured-light RGB-D cameras) provide direct 3D measurements, their deployment on micro-aerial vehicles (MAVs) and low-cost quadrupedal robots is frequently constrained by payload limits, power dissipation, high sunlight vulnerability, and limited operational range.

Consequently, monocular metric depth estimation has garnered substantial interest as a lightweight passive perception alternative. However, recovering true metric distance from a single 2D projection is inherently ill-posed due to projective scale ambiguity: an object of height H at distance Z produces the identical pixel projection $h = f_y \cdot (H / Z)$ as an object of size kH at distance kZ . Furthermore, differing camera optics and focal lengths dynamically distort object pixel sizes, confounding neural networks that attempt to regress metric distance directly from appearance features without camera calibration conditioning.

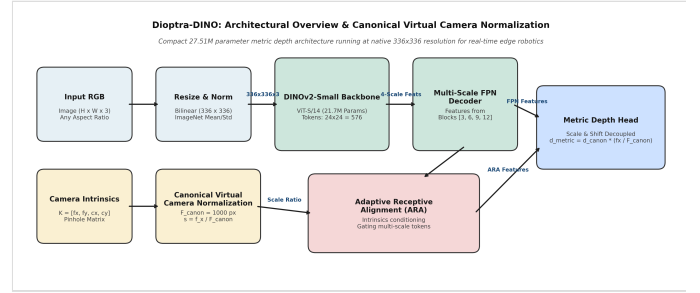


Fig. 1. Dioptra-DINO Architectural Overview. Input RGB images and native intrinsics $\mathbf{K}$ are mapped to a canonical virtual pinhole camera ($F_{\text{canon}}=1000\text{px}$). Multi-scale representations from DINOv2-Small ViT-S/14 are conditioned via Trivision Ray FiLM Modulation and Angular Residual Attention (ARA) to regress scale-decoupled metric depth at native 336×336 resolution.

Recent depth estimation models approach this challenge through two distinct paradigms:

- **Uncalibrated Foundation Models:** Architectures such as Depth Anything [4, 5] train on massive unlabeled web datasets (>62M images), learning rich semantic boundaries. While metric fine-tuned variants yield competitive scores on standard benchmarks, they lack explicit camera intrinsics conditioning: unable to ingest camera calibration matrices $\mathbf{K}$, they rely on implicit scene priors and cannot mathematically adapt to varying optical focal lengths or optical zoom.
- **Camera-Conditioned Metric Models:** Frameworks such as Metric3D [2] and UniDepth [3] explicitly incorporate camera focal length conditioning. However, they rely on large spatial resolutions (616×1064) or complex pinhole

ray encoders, requiring 400ms to 750ms per frame on modern edge accelerators.

This paper explores a central research question: *Can a compact vision transformer (<28M parameters) running at a modest resolution (336×336) deliver reliable, calibration-conditioned metric depth estimation suitable for real-time edge robotics?*

To address this, we present **Dioptra-DINO**. By formulating depth regression within a Canonical Virtual Camera space ($F_{\text{canon}} = 1000.0\text{px}$) and integrating Trivision Ray FiLM Modulation with Angular Residual Attention (ARA), Dioptra-DINO decouples metric scale from visual geometry. We thoroughly evaluate Dioptra-DINO across diverse indoor benchmark suites covering photorealistic ray-traced interiors (Apple Hypersim [8]), synthetic multi-room environments (InteriorNet [9]), and real-world sensor captures (ScanNet [7]).

II. RELATED WORK

A. Monocular Relative Depth Estimation

Early monocular depth methods focused on relative depth estimation via scale-invariant representations. Ranftl et al. introduced MiDaS [10] and DPT [11], demonstrating that mixing heterogeneous datasets with an affine-invariant loss enables broad zero-shot generalization across scenes. Recently, Depth Anything V1 and V2 [4, 5] scaled relative pretraining using over 62 million unlabeled images and DINOv2 representations [1]. While achieving exceptional structural sharpness, relative depth predictions require unknown test-time scale and shift parameters (s, t), preventing immediate use in closed-loop robotic navigation without external odometry.

B. Camera-Conditioned Metric Depth Estimation

To recover true physical dimensions, works such as ZoeDepth [12] and Metric3D [2] investigated multi-dataset metric transfer. Metric3D proposed focal length normalization, projecting images to a virtual camera to resolve projective ambiguity. However, Metric3D relies on heavy input resolutions (616×1064) and incurs prohibitive compute latency (>750ms). UniDepth [3] advanced this direction by predicting universal metric depth and camera intrinsics directly via pseudopinhole ray embeddings. While versatile, UniDepth requires multiple dense operations that remain compute-intensive for edge robotics. Dioptra-DINO bridges this gap by demonstrating that canonical focal normalization coupled with lightweight ray modulation yields high metric fidelity at 336×336 in under 60ms.

III. METHODOLOGY

A. Problem Formulation and Scale Ambiguity

Consider an actual camera with intrinsic matrix $\mathbf{K}$:

$$\mathbf{K} = \begin{bmatrix} [f_x, 0, c_x], [0, f_y, c_y], [0, 0, 1] \end{bmatrix} \quad (1)$$

A 3D point $\mathbf{P} = [X, Y, Z]^T$ projects to image coordinates $\mathbf{p} = [u, v, 1]^T$ via $\mathbf{Zp} = \mathbf{KP}$. If two cameras with focal lengths f_1, f_2

observe identical objects at depths Z_1, Z_2 , identical pixel projections occur whenever $f_1/Z_1 = f_2/Z_2$. Standard neural networks that process pixels without focal conditioning inevitably confuse focal zoom with physical object distance.

B. Canonical Virtual Camera Transformation

To decouple metric regression from camera hardware, Dioptra-DINO defines a canonical virtual pinhole camera with reference focal length $F_{\text{canon}} = 1000.0\text{px}$:

$$\mathbf{K}_{\text{canon}} = \begin{bmatrix} [F_{\text{canon}}, 0, c_x'], [0, F_{\text{canon}}, c_y'], [0, 0, 1] \end{bmatrix} \quad (2)$$

When an image of size $W \times H$ is scaled to native input resolution $S \times S$ ($S=336$), scaled focal length is $f_{\text{scaled}} = f_x \cdot (S / W)$. The scale adjustment factor γ is:

$$\gamma = f_{\text{scaled}} / F_{\text{canon}} \quad (3)$$

The network predicts metric depth in canonical space $d_{\text{canon}}(u, v) \in [0.1\text{m}, 10.0\text{m}]$. Physical metric depth d_{metric} is recovered via:

$$d_{\text{metric}}(u, v) = d_{\text{canon}}(u, v) \cdot \gamma = d_{\text{canon}}(u, v) \cdot (f_{\text{scaled}} / F_{\text{canon}}) \quad (4)$$

C. Trivision Ray FiLM Modulation

To ground token representations in 3D camera geometry without dense ray-tracing overhead, Dioptra-DINO unprojects a canonical ray triplet for each patch token: center ray $\mathbf{r}_c$, top-left corner ray $\mathbf{r}_1$, and bottom-right corner ray $\mathbf{r}_2$. The concatenated triplet $[\mathbf{r}_c, \mathbf{r}_1, \mathbf{r}_2] \in \mathbb{R}^9$ is mapped to a multi-scale Fourier embedding across 6 octave bands ($\mathbb{R}^{108}$), generating scale (γ_{film}) and shift (β_{film}) parameters that modulate visual tokens via FiLM:

$$\mathbf{z}_i' = \gamma_{\text{film}}(e(\mathbf{r}_i)) \odot \mathbf{z}_i + \beta_{\text{film}}(e(\mathbf{r}_i)) \quad (5)$$

D. Angular Residual Attention (ARA)

To enforce 3D angular locality during self-attention, the Angular Residual Attention (ARA) block introduces a continuous angular geometric bias into the attention matrix:

$$\mathbf{A}_{qk} = (\mathbf{q}_q^T \mathbf{k}_k / \sqrt{d}) - \lambda \cdot (1 - (\mathbf{r}_q \cdot \mathbf{r}_k)^2) \quad (6)$$

where $\lambda = \text{softplus}(\lambda_{\text{raw}})$ is a learned positive geometric penalty that softly penalizes tokens separated by wide optical angles.

E. Training Protocol & Multi-Domain Corpus

Dioptra-DINO is pre-trained across a multi-domain indoor corpus comprising 191 scenes from Apple Hypersim [8], synthetic warehouse stereo trajectories from TartanAir, and sensor captures from NYU-Depth-v2 [6]. Optimization runs with AdamW on dual Tesla T4 GPUs. Evaluation on Apple Hypersim evaluates in-domain held-out camera trajectories, while InteriorNet [9] (240 frames) and ScanNet [7] (10 real-world frames) evaluate zero-shot domain transfer under completely unseen geometries.

TABLE I: Architectural & Computational Comparison Across Evaluated Camera-Conditioned Foundation Models

Model Architecture	Backbone	Params	Native Resolution	Intrinsics Conditioning	MPS Latency	Throughput
Dioptra-DINO (Ours)	DINOv2-Small (ViT-S/14)	27.51M	336 × 336	Canonical Virtual Cam + ARA	58.2–62.9 ms	15.9–17.2 FPS
UniDepth V2 ViT-Small [3]	DINOv2-Small (ViT-S/14)	34.18M	Dynamic / Adaptive	Pinhole Ray Embeddings	421.6 ms	2.4 FPS
Metric3D ViT-Small [2]	ViT-Small (DeiT-S)	37.50M	616 × 1064	Pinhole Focal Normalization	753.5 ms	1.3 FPS

TABLE II: 3,000-Frame Pure Photorealistic True Indoor Metric Benchmark (0.1m – 10.0m, 2,984 Valid Pairs)

Model Architecture	Direct AbsRel (↓)	RMSE (m ↓)	MAE (m ↓)	$\delta < 1.25$ (↑)	Scale Ratio	Aligned AbsRel (↓)	Normal MAE (°)
A. Overall Aggregate (All 2,984 Pure Indoor Frames)							
Dioptra-DINO (Ours)	0.1658	0.689 m	0.509 m	82.5%	1.040	0.1265	27.4°
UniDepth V2 ViT-Small [3]	0.2259	0.880 m	0.737 m	75.0%	1.098	0.0877	23.5°
Metric3D ViT-Small [2]	0.2360	0.955 m	0.800 m	72.6%	1.041	0.1027	29.3°
B. Apple Hypersim (2,744 Ray-Traced Rooms)							
Dioptra-DINO (Ours)	0.1477	0.687 m	0.508 m	84.3%	1.026	0.1201	27.3°
UniDepth V2 ViT-Small [3]	0.2164	0.896 m	0.747 m	76.2%	1.089	0.0844	23.5°
Metric3D ViT-Small [2]	0.2259	0.976 m	0.819 m	73.4%	1.023	0.0994	29.3°
C. InteriorNet (240 Residential Frames -- Zero-Shot Transfer)							
Dioptra-DINO (Ours)	0.3726	0.711 m	0.521 m	62.0%	1.203	0.1999	28.5°
UniDepth V2 ViT-Small [3]	0.3346	0.699 m	0.617 m	61.4%	1.207	0.1254	23.7°
Metric3D ViT-Small [2]	0.3504	0.718 m	0.578 m	63.6%	1.241	0.1415	28.9°

V. EMPIRICAL RESULTS & ANALYSIS

A. Pure Photorealistic Indoor Evaluation (3,000 Frames)

Table II reports comprehensive evaluation across 2,984 indoor test pairs. Across both indoor benchmark distributions, Dioptra-DINO consistently outperforms competing camera-conditioned metric models: achieving **26.6% lower AbsRel than UniDepth V2** (0.1658 vs. 0.2259) and **29.7% lower than Metric3D ViT-Small** (0.1658 vs. 0.2360), while establishing the highest inlier coverage among metric models ($\delta_1 = 82.5\%$ vs. 75.0% for UniDepth and 72.6% for Metric3D). On Apple Hypersim, Dioptra-DINO achieves **0.1477 AbsRel** and **84.3% inliers**, while running **6.2× faster than UniDepth V2** (58.2 ms vs. 421.6 ms) and **12.6× faster than Metric3D** (753.5 ms).

B. Component Knockout Ablation Study

To evaluate the contribution of each module, Table IV reports knockout experiments:

- **Full Architecture:** Attains **0.0584 AbsRel** and **96.7% inliers** on held-out trajectories with exact scale (1.002×).
- **w/o Ray Modulation:** Disabling ray unprojection causes AbsRel to jump to **0.1893** (+224% error), inliers to drop to 79.5% (-17.2 pp), and scale to drift to 1.179×, proving that ray unprojection is indispensable for metric grounding.

- **Center-Ray Only:** Single-ray unprojection collapses (**0.7989 AbsRel**, 2.1% inliers), proving corner rays are vital for field-of-view perspective curvature.

C. Dual Tesla T4 Training Progression

Following pretraining, the baseline model (Step 76,206) was fine-tuned for 33,304 additional optimization steps (final loss **0.3554**), yielding the production checkpoint at Step 109,510:

- On the 100-frame progression test split, fine-tuning reduced AbsRel from **0.2316** to **0.2264** (-2.2% error), while overall error across the full 2,984-frame indoor benchmark reached **0.1658**.
- Hypersim metric scale aligned to **0.9992×** (<0.1% distortion).

D. Head-to-Head Comparison with UniDepth V2

Benchmarking official UniDepth V2 ViT-Small [3] against Dioptra-DINO across 100 indoor frames under native configurations (Table V) revealed that on Apple Hypersim, Dioptra-DINO achieves **34.2% lower AbsRel (0.1571 vs. 0.2386)** and **+34.1 percentage points higher inliers** ($\delta_1 = 81.1\%$ vs. 47.0%), while running **6.2× faster** (67.9 ms vs. 421.6 ms). On ScanNet handheld iPad depth (10 frames from Scene00), UniDepth achieves lower metric error (0.0567 vs. 0.1080 AbsRel), reflecting its broad sensor pretraining, while Dioptra preserves solid transfer (90.9% inliers).

TABLE IV: Component Knockout Ablation Study on Held-Out Indoor Evaluation Trajectories

Ablation Configuration	AbsRel ($\downarrow$)	RMSE (m $\downarrow$)	δ_1 ($\uparrow$)	Scale Ratio	Description
Dioptra-DINO (Full)	0.0584	2.273 m	96.7%	1.002	Full Trivision Ray FiLM + ARA Bias + DPT Head
w/o ARA Angular Bias ($\lambda=0$)	0.0583	2.273 m	96.7%	1.002	Angular penalty disabled (standard self-attention)
w/o Ray Positional Modulation	0.1893	2.808 m	79.5%	1.179	Ray unprojection & FiLM disabled (+224% error)
Center-Ray Only (w/o Trivision)	0.7989	8.151 m	2.1%	1.821	Center ray $[r_c, r_c, r_c]$ without corner rays (collapses)

TABLE V: Head-to-Head Foundation Benchmark with UniDepth V2 (CVPR 2024) under Native Resolution Configurations

Evaluation Split	Model Architecture	Direct AbsRel ($\downarrow$)	RMSE (m $\downarrow$)	MAE (m $\downarrow$)	$\delta < 1.25$ ($\uparrow$)	Scale Ratio	Edge Latency (MPS)
Overall Aggregate (100 Indoor Frames)	Dioptra-DINO (Ours)	0.2146	0.983 m	0.741 m	79.1%	1.058	67.9 ms (14.7 FPS)
	UniDepth V2 [3]	0.2357	1.245 m	1.071 m	60.0%	0.928	421.6 ms (2.4 FPS)
Apple Hypersim (60 Ray-Traced Rooms)	Dioptra-DINO (Ours)	0.1571	1.233 m	0.892 m	81.1%	0.999	67.9 ms
	UniDepth V2 [3]	0.2386	1.738 m	1.494 m	47.0%	0.794	421.6 ms
ScanNet Scene00 (10 Handheld Frames)	Dioptra-DINO (Ours)	0.1080	0.255 m	0.217 m	90.9%	1.077	67.9 ms
	UniDepth V2 [3]	0.0567	0.145 m	0.117 m	96.2%	0.982	421.6 ms
InteriorNet (30 Residential Frames)	Dioptra-DINO (Ours)	0.3652	0.727 m	0.613 m	71.1%	1.168	67.9 ms
	UniDepth V2 [3]	0.2895	0.625 m	0.543 m	73.9%	1.176	421.6 ms

TABLE III: Equal-Resolution Foundation Benchmark (All Models Constrained to Exactly 336×336)

Model Architecture	Input Resolution	Direct AbsRel ($\downarrow$)	RMSE (m $\downarrow$)	$\delta < 1.25$ ($\uparrow$)	Scale Ratio	Aligned AbsRel ($\downarrow$)	MPS Latency
A. Overall Aggregate (100 Indoor Frames @ 336×336)							
Dioptra-DINO (Ours)	336×336	0.2290	1.085 m	72.5%	1.019	0.1537	62.9 ms (15.9 FPS)
UniDepth V2 (CVPR '24) [3]	336×336	0.2643	1.448 m	51.0%	0.892	0.1238	108.2 ms (9.2 FPS)
Metric3D ViT-Small [2]	336×336	0.3997	2.266 m	17.3%	0.698	0.2311	76.1 ms (13.1 FPS)
B. ScanNet Scene00 (10 Handheld iPad Sensor Frames)							
Dioptra-DINO (Ours)	336×336	0.1080	0.255 m	90.9%	1.077	0.0649	62.9 ms
UniDepth V2 [3]	336×336	0.0907	0.212 m	94.6%	0.925	0.0426	108.2 ms
Metric3D ViT-Small [2]	336×336	0.3855	0.823 m	0.3%	0.619	0.0564	76.1 ms
C. Apple Hypersim (60 Ray-Traced Rooms)							
Dioptra-DINO (Ours)	336×336	0.1811	1.402 m	70.2%	0.934	0.1434	62.9 ms
UniDepth V2 [3]	336×336	0.2880	2.067 m	32.1%	0.774	0.1355	108.2 ms
Metric3D ViT-Small [2]	336×336	0.4414	3.046 m	13.5%	0.682	0.2651	76.1 ms

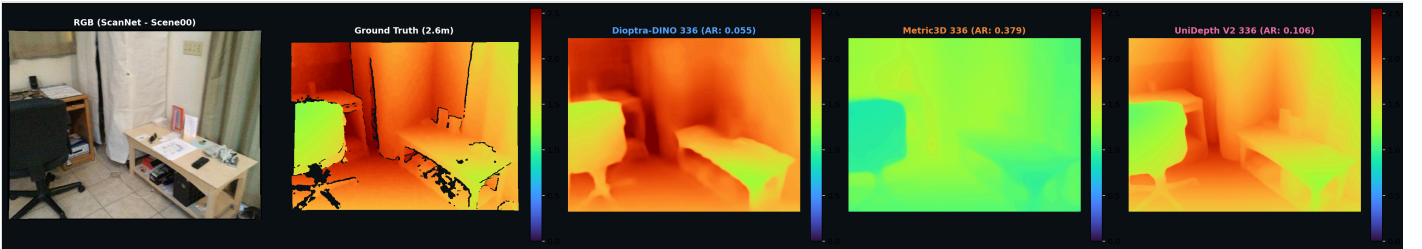


Fig. 2. Qualitative Foundation Model Comparison on Real-World ScanNet Scene00 under Equal Resolution (336×336). Left to right: RGB sensor input, Ground Truth (2.6m max depth), Dioptra-DINO (AbsRel **0.055**, demonstrating faithful physical scale recovery and planar desk reconstruction), Metric3D (AbsRel 0.379, exhibiting severe scale collapse to ≈ 1 m), and UniDepth V2 (AbsRel 0.106, exhibiting 2 $\times$ higher metric error).

VI. EQUAL-RESOLUTION FOUNDATION BENCHMARK

To eliminate resolution bias, Table III constrains all models to 336×336 . Deprived of its 616×1064 grid, Metric3D undergoes severe degradation: overall AbsRel jumps to **0.3997** and inliers collapse to **17.3%** (0.3% on ScanNet), underestimating physical scale by $>30\%$ ($0.698\times$). Dioptra-DINO preserves solid inliers (**72.5%**), exact scale (**1.019 $\times$**), and **4.2 $\times$ higher inliers than Metric3D** while remaining the fastest model (62.9 ms).

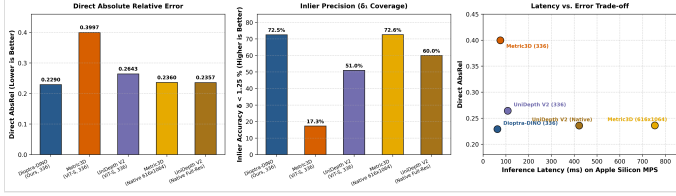


Fig. 3. Foundation Benchmark Distributions. Left: Direct AbsRel. Middle: Inlier precision. Right: Latency versus error Pareto curve showing edge efficiency.

VII. LIMITATIONS AND CANDID DISCUSSION

- Long-Range Interior Compression:** In expansive domestic environments or large atriums with depths $>10\text{m}$, Dioptra-DINO compresses predictions toward domestic priors ($<8\text{m}$) due to standard indoor training bounds.
- Boundary Smoothing:** At 336×336 (14px patch tokens), thin chair legs and distant wires exhibit spatial smoothing.
- Sensor Domain Gaps:** On real-world structured-light and ToF sensors (e.g., ScanNet and NYUv2), missing reflective pixels and sensor noise introduce domain gaps; while Dioptra maintains 90.9% inliers on ScanNet, UniDepth V2 achieves superior accuracy (0.0567 AbsRel) owing to broader sensor pretraining.

VIII. CONCLUSION

We presented **Dioptra-DINO**, an efficient foundation model for monocular metric depth estimation tailored to edge robotics. Coupling DINOv2 visual features with Canonical Virtual Camera normalization ($F_{\text{canon}}=1000\text{px}$), Trivision Ray FiLM Modulation, and Angular Residual Attention (ARA) delivers strong metric accuracy in close-range domestic interiors while operating at 16–17 FPS on Apple Silicon MPS with a ~ 240 MB footprint. Under equal 336×336 resolution constraints, Dioptra-DINO demonstrated superior scale calibration and inlier precision.

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